

2026 AEA Convention & Trade Show



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Classifying Alterations for In-Service Aircraft

Regulatory Framework, ASTM F3361, and FAA Compliance



Why This Matters

- Inconsistent classification across industry
- Increased FAA scrutiny on data approval
- Risk of improper return to service
- SMS integration and hazard identification



The Core Challenge

- Regulations define WHAT is major/minor
- They do not define HOW to determine it



Regulatory Framework

- 14 CFR Part 21.93 – Changes in Type Design
- 14 CFR Part 1.1 – Definitions
- 14 CFR Part 43 Appendix A – Major Alterations



Three-Tier In-Service Modification System

Change/Modification	Data Requirements
Major Change in Type Design (14 CFR §21.93)	STC/AMT
Major Alteration (14 CFR part 1/part 43 appendix A)	Approved Data / Data Approval
Minor Alteration (14 CFR Part 1)	Acceptable Data



Key Distinction

Major Change in Type Design ≠ Major Alteration



- Provides process for data approval
- Applies to major repairs and alterations
- Does NOT determine major vs minor



F3361: Standard Guide for Classifying Alterations for In-Service Aircraft under FAA Authority Oversight



Designation: F3361 – 18

Standard Guide for Classifying Alterations for In-Service Aircraft under FAA Authority Oversight¹

This standard is issued under the fixed designation F3361; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

1. Scope

1.1 This guide is intended for Part 23 and predecessor aircraft (see Civil Air Regulations (CAR) 3) and:

1.1.1 Is applicable to aircraft to which Part 43 applies.

1.1.2 Addresses both initial installation and replacement of aircraft articles.

1.1.3 Is a guide for classifying an alteration.

1.1.4 The intended audience of this guide is individuals who have been tasked with evaluating alterations for the purpose of categorizing an alteration for the purpose of satisfying the regulations.

1.1.5 Is an alteration to an article.

14 CFR Aeronautics and Space, Vol 1, Chapter I, Subchapter C, Part 23 Airworthiness Standards: Normal, Utility, Acrobatic, and Commuter Category Airplanes

14 CFR Aeronautics and Space, Vol 1, Chapter I, Subchapter C, Part 43 Maintenance, Preventive Maintenance, Rebuilding, and Alteration, including Amendments 43-1 through 43-49

14 CFR Aeronautics and Space, Vol 2, Chapter I, Subchapter D, Part 65 Certification: Airmen Other Than Flight Crewmembers

14 CFR Aeronautics and Space, Vol 3, Chapter I, Subchapter H, Part 145 Repair Stations

2.2 FAA Standards³



ASTM F3361

- Accepted means of compliance by FAA
- Provides structured evaluation method



ASTM F3361 provides a checklist!



F3361 – 18

Aircraft Make _____

Aircraft SN _____

Aircraft Model _____

Aircraft N _____

Date _____

Work Order _____

Completed by _____

Describe Alteration: _____

Evaluating the Modification for Change in Type Design (14 CFR 21.93)

Does the proposed alteration have an appreciable effect on the certificated weight? (for example, a change in the maximum takeoff weight limitations, minimum landing weight limitations, etc.)

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on the certificated balance? (for example, a change in the forward or aft center of gravity limits, etc.)

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on the structural strength?

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on the performance?

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on the powerplant operation?

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on the flight characteristics?

Yes: _____

No: _____

Why?: _____

Does the proposed alteration have an appreciable effect on other characteristics affecting the airworthiness?

Yes: _____

No: _____

Why?: _____



Six Evaluation Criteria

Weight

Balance

Structural
Strength

Reliability

Operational
Characteristics

Airworthiness



Major Change in Type Design (§ 21.95)

- Does the proposed alteration have an appreciable effect on the:
 - [certificated] **weight** (of the type certificated product)?
 - [certificated] **balance** (of the type certificated product)?

Yes to any: This is a major change to type design and requires an application for a supplemental or amended type certificate.

No to all: Continue.



Appreciable Change

- a change that is large enough to be noticed using an appropriate unit of measure.
- *With modern technology, all changes could be measured, i.e. increased weight, thickness, dimensions, etc., which is not the intent of this definition.*



- 6.2.2.1 Certified Weight and Balance—Examples of alterations that may appreciably affect the Certified Weight and Balance include, but are not limited to:
 - (1) Changes that increase the certificated maximum weight limits (increases in the maximum gross weight, maximum take-off, or landing weights).
 - (2) Changes in the certificated center of gravity range limits (for example, decreasing the forward limit or increasing the aft limit).



Major Change in Type Design (§ 21.95)

- Does the proposed alteration have an appreciable effect on the:
 - **structural strength** (of the type certificated product)?

Yes to any: This is a major change to type design and requires an application for a supplemental or amended type certificate.

No to all: Continue.



- 6.2.2.2 Structural Strength—Examples of alterations that may appreciably affect structural strength include, but are not limited to:
 - (1) Changes to primary structures (structure that carries flight, ground, or pressure loads as defined in AC 25.571-1).
 - (2) Substituting an engine, propeller, rotor, or airframe primary structural materials (such as replacing a reciprocating engine with a turbine engine or increasing horsepower output by 10 % or more).



Major Change in Type Design (§ 21.95)

- Does the proposed alteration have an appreciable effect on the:
 - **reliability** (of the type certificated product)?

Yes to any: This is a major change to type design and requires an application for a supplemental or amended type certificate.

No to all: Continue.



- 6.2.2.3 Reliability—Examples of alterations that may appreciably affect reliability include, but are not limited to:
 - (1) Changes to manifolding, air induction systems or air intake doors, engine cowling or baffle that affect the flow of engine cooling air and carburetor/fire ignition heat rises.
 - (2) Changing the basic engine or propeller design, controls, and operating limitations.
 - (3) Changes that include engine/propeller adjustments and setting limitations that affect power output.



- 6.2.2.3 Reliability—Examples of alterations that may appreciably affect reliability include, but are not limited to:
 - (4) Modifications to approved avionics equipment that affect reliability or airworthiness, such as changes:
 - (a) Deviating from the design environment performance.
 - (b) Deviating from the component manufacturers operating limitations.
 - (c) To software. (Note that Changes to Software refers to modification of the code, not of doing an OEM approved software update or revision.)
 - (d) To wire shielding that may affect High Intensity Radiated Fields (HIRF) and Electromagnetic Interference (EMI).



Major Change in Type Design (§ 21.95)

- Does the proposed alteration have an appreciable effect on the:
 - **operational characteristics**
(of the type certificated product)?

Yes to any: This is a major change to type design and requires an application for a supplemental or amended type certificate.

No to all: Continue.



- 6.2.2.4 Operational Characteristics—Examples of alterations that may appreciably affect operational characteristics include, but are not limited to:
 - (1) Changes or relocation of systems (including hydraulic, oil, and fuel systems) and equipment that affects structural integrity, flight, ground handling characteristics, or noise/acoustics of the aircraft.
 - (2) Changes that alter the movable control surfaces that affect the dynamic or static balance, or both, alter the aerodynamic contour of movable control surfaces, or change the weight distribution.
 - (3) Changes in control surface travel, control system mechanical advantage, location of control system component parts, or direction of motion.



- 6.2.2.4 Operational Characteristics—Examples of alteration that may appreciably affect operational characteristics include, but are not limited to:
 - (4) Changes in basic dimensions or external aerodynamic contour/configuration of the aircraft such as wing and tail planform or incidence angles, canopy, cowlings, contour or radii, the location of wing and tail fairings, winglets, wing lift struts, tiptanks, windows, and doors.
 - (5) Installation of support structure for appliances, or support structure and appliances, to the exterior (that is, night sun beacon, camera, spray/dusting equipment) on rotorcraft only.
 - (6) Changes to flight-critical electrical/electronic systems such as electronic flight controls or the engine control system, Full Authority Digital Engine Control (FADEC), fly by wire, and so forth.



- 6.2.2.4 Operational Characteristics—Examples of alterations that may appreciably affect operational characteristics include, but are not limited to:
 - (7) Changes that affect aircraft performance, drag, engine power, revolutions per minute (r/min), or exhaust muffler.
 - (8) Changes affecting noise or flight characteristics.
 - (9) Rotorcraft items, such as external search lights, skis, baskets, and so forth.
 - (10) Changes that increase the operational limits (maximum speed limits such as VA, VFE, VNE; minimum speed limitations such as stall speed; increases in service ceiling, and so forth).



Major Change in Type Design (§ 21.95)

- Does the proposed alteration have an appreciable effect on the:
 - **characteristics affecting the airworthiness**
(of the type certificated product)?

Yes to any: This is a major change to type design and requires an application for a supplemental or amended type certificate.

No to all: Continue.



- 6.2.2.5 Airworthiness—Examples of alterations that may appreciably affect the airworthiness include, but are not limited to:
 - (1) Changes to landing gear and related components, such as internal parts of shock struts, length, geometry of members, changes to brake and brake systems, or additions.
 - (2) Changes to systems that affect aircraft airworthiness, such as:
 - (a) Relocation of exterior fuel vents or battery vents.
 - (b) Crew or passenger liquid oxygen (LOX) or on-board generating systems.
 - (c) External critical access doors, Auxiliary Power Unit (APU) ram air, nacelle blowout doors, fuel drain.
 - (3) Major deviations to STCs.



- 6.2.2.5 Airworthiness—Examples of alterations that may appreciably affect the airworthiness include, but are not limited to:
 - (4) Changes to oil, hydraulic, pneumatic, and fuel lines or systems that affect their operation or installation and flammability requirements, such as:
 - (a) New types of hoses or hose fittings, or both, that may not meet installation requirements such as flow rate and flammability requirements.
 - (b) Changes to fuel dump valves.
 - (c) New oil/fuel/hydraulic line materials or sealants.
 - (d) Change to, or addition of, permanent fuel tanks or fuel system components.



- 6.2.2.5 Airworthiness—Examples of alterations that may appreciably affect the airworthiness include, but are not limited to:
 - (5) Changes in fixed fire extinguisher or detector systems that affect system effectiveness or reliability, such as:
 - (a) Relocation of discharge nozzles, detector units, or fixed fire extinguisher bottles.
 - (b) Using new or different detector components.
 - (c) Decreasing the amount or changing the type of extinguishing agents.



- 6.2.2.5 Airworthiness—Examples of alterations that may appreciably affect the airworthiness include, but are not limited to:
 - (6) Changes that include the substitution of engine/APU/propeller/airframe materials that affect structural integrity, lightning protection, flight characteristics, or noise/acoustics.
 - (7) Any other complex special process that, if not properly performed, has a significant adverse effect on the integrity of the product.
 - (8) Major alterations to propellers.



- 6.2.2.6 Crashworthiness—Examples of alterations that may appreciably affect crashworthiness include, but are not limited to:
 - (1) Changes to the aircraft structure, cabin interiors, or equipment relocation.
 - (2) Changes that increase the certificated seating capacity, excluding sport parachute jumping configuration.
 - (3) Changes that include the substitution of engine/propeller/airframe materials that affect fire protection, lightning protection, or flammability.



IF: Major Change in Type Design

- AC 21-40A - Guide for Obtaining a Supplemental Type Certificate
- FAA Order 8110.4C - Type Certification



Major Alterations (§ 1.1)

- Is it possible that the proposed alteration *might* have an appreciable effect on the:
 - [certificated] weight?
 - [certificated] balance?
 - structural strength?
 - performance?
 - powerplant operation?
 - flight characteristics?
 - characteristics affecting the airworthiness?

Yes or maybe to any: The proposed change is a major alteration requiring approved data, recording of a FAA Form 377 and a log book entry.

Confirmed no to all: Continue.



What does this mean?

- might have an appreciable effect



What does this mean?

- might have an appreciable effect

We Don't Know....



Major Alterations

(14 CFR Part 43 Appendix A(a)(1) Major airframe alterations)

- Airframe major alterations.
 - Alterations of the following parts and alterations of the following types, **when not listed in the aircraft specifications issue by the FAA**, are airframe major alterations:



In Service Modifications: 3 Tiers

Change/Modification	Measurable Affect on 6 Characteristics
Major Change in Type Design (21.93)	
Major Alteration (part 1)	
Minor Alteration (part 1)	



In Service Modifications: 3 Tiers

Change/Modification	Measurable Affect on 6 Characteristics
Major Change in Type Design (21.93)	Confirmed YES
Major Alteration (part 1)	
Minor Alteration (part 1)	



In Service Modifications: 3 Tiers

Change/Modification	Measurable Affect on 6 Characteristics
Major Change in Type Design (21.93)	Confirmed YES
Major Alteration (part 1)	
Minor Alteration (part 1)	Confirmed NO



In Service Modifications: 3 Tiers

Change/Modification	Measurable Affect on 6 Characteristics
Major Change in Type Design (21.93)	Confirmed YES
Major Alteration (part 1)	Maybe ????
Minor Alteration (part 1)	Confirmed NO



Avionics Focus

- EMI/HIRF
- Software changes
- Environmental qualification impacts



If You Don't Know

- Treat as Major Alteration



Data Requirements

Change/Modification	Data Requirements
Major Change in Type Design	STC/AMT
Major Alteration	Approved Data / Data Approval
Minor Alteration	Acceptable Data



Documentation

- Document evaluation
- Attach to work order
- Logbook entry per §43.9
- Identify major/minor



Common Errors

- Using AC 43-210A to classify
- Skipping documentation
- Classifying by convenience



AEA Position

- Promote ASTM F3361
- Improve consistency
- Support FAA oversight



SMS Integration

- Supports hazard identification
- Aligns with risk-based decision making



Summary

- Regulations define outcomes
- ASTM defines method
- Documentation ensures compliance



Questions



Alteration Classification Flowchart

